Inorganic Cume **February 28, 1998**

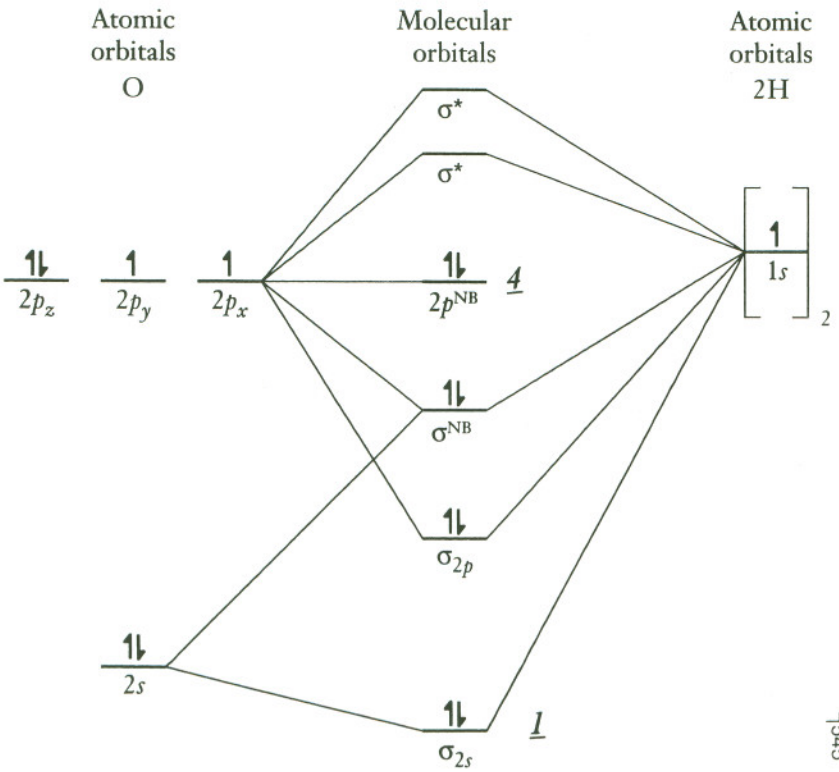
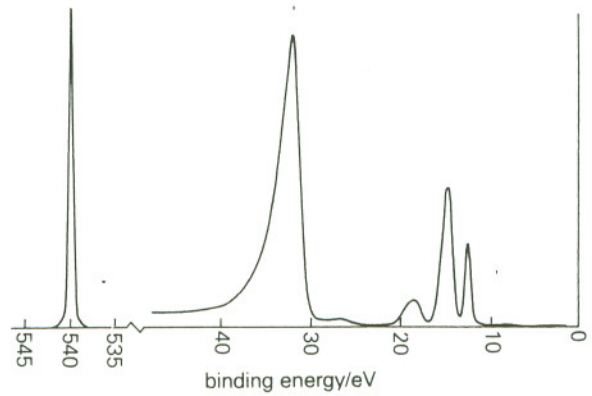
1) An approximate MO diagram for the H_2O molecule is shown below. Which atomic orbitals are the major contributors to the levels labeled *1* and *4* ?

2) The photoelectron spectrum for the H_2O molecule is shown below. Briefly, account for the five peaks observed. [*Note the "break" on the binding energy axis.*]
Can you rationalize this spectrum using the traditional localized valence bond model for water?

3). What molecular phenomenon leads to the broadening of the peaks for the four lower binding energies?

4) A recent seminar speaker(J. Kolis) dealt with reactions in *supercritical* water (scH_2O). What is a supercritical fluid and how is it produced ?

5) Discuss some of the differences between scH_2O as a solvent and $H_2O_{(l)}$ for chemical reactions. Include some of the advantages and disadvantages of scH_2O as reaction medium in your answer.

6) Many of the reactions described by Professor Kolis involved the addition of a *mineralizer* (e.g. OH^-, Cl^-, etc.). What function did these species serve in the synthetic procedures?

7) Professor Kolis emphasized that the products obtained in many of the reactions he studied were *kinetically controlled* . What did he mean and why do you think he emphasized this point?